The fdamador neutrino: a report

Nov 4, 2026 · @FDA

Summary

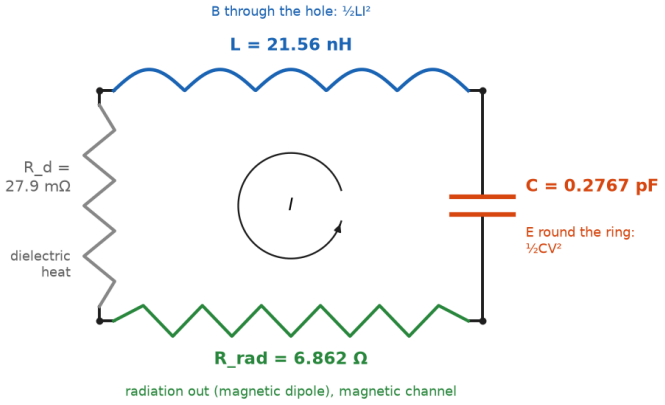
Turning the toroidal photon's field inside out gives a neutral, self-sustaining ring of electric field, which matches two neutrino properties but cannot stay coherent long enough to be one. Hypothesis H28 is recorded as *bounded*.

- **What matches:** the inverted ring has zero charge and zero time-averaged magnetic moment, and its crossed circuit form keeps the half-integer (spin ½) modes.
- **What fails:** in vacuum it radiates away in about 3×10^{-16} s. SN1987A neutrinos survived about 5×10^{4} s of their own time, a gap of about 20 orders of magnitude.
- **The inflow:** the zero-point field of its own radiation resistance balances the outflow. The ring holds $\frac{1}{2}\hbar\omega_0$ forever with the energy ledger closed, but as random noise, not as a particle.
- **Missing entirely:** mass and the weak interaction, which is how every neutrino has been detected.
- **Implosion and explosion:** the ring takes in the exact time reverse of its ring-down whole (0.99966) and explodes it back out, but every implosion is paid for by an incoming wave from outside.
- **Flavours (H39):** travelling, standing and cancelling waves on the ring are stable and, with a defect, convert in the form of neutrino oscillation; tuned, they match the measured mixing angles, but the three types span only two states and carry no mass or weak interaction.
- **A spinning ring (H40):** turning the ring gives a CP phase and T violation in form, but only at 11 to 444 times the burst speed of the strongest fibre; it gives no left-handedness, spin ½, mass or confinement.
- **Flavours as three spin states (H41):** turning one way, not turning, turning the other way are three genuinely independent states, but switching between them needs a torque from outside, the measured mixing is reached only by fitting, and outside rotation cannot supply the mass at any reachable speed.
- **Two neutrinos (H42):** two rings couple as resonators; the antiphase pair radiates about 22× more slowly than one ring and a chain compounds as N^2.92, but reaching the needed lifetime takes about a million phase-locked rings, and the model rings would interact about $10^{9}\times$ more strongly than real neutrinos do.

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As a circuit, the model is a series RLC loop. The E field round the ring is displacement current through the dielectric, so the ring is a capacitor (C = 0.2767 pF, storing ½CV²). The B field through the hole is the ring's self-inductance (L = 21.56 nH, storing ½LI²). The radiation resistance R_rad = 6.862 Ω carries energy out as magnetic-dipole radiation, and R_d = 27.9 mΩ is the dielectric's heat. Energy moves from C to L and back every quarter period, at f0 = 1/(2π√(LC)) = 2.0606 GHz with Q = 40.5. This is the circuit equivalent of the model, not of the measured neutrino: it has no mass and no weak interaction.

The fdamador neutrino as a circuit: a series RLC loop (bench ring, a = 10 mm, εr = 100)



Resonance	f0 = 1/(2π√(LC)) = 2.0606 GHz
Quality	Q = ω0L/(R_rad + R_d) = 40.51 (radiation alone 40.68)
Period, decay	T = 0.485 ns; energy time Q/ω0 = 3.13 ns
Energy per cycle	E and B trade it each quarter period; 14 % leaves per cycle
Where from	L: Maxwell, Treatise Art. 706, μ0a(ln(8a/r) − 7/4) R_rad: (η0π/6)(ka)⁴ (Balanis 5-24), ka = 0.432 C: set by the ring's εr = 100 so that f0 is the TE01δ mode
Checked	kronspice AC peak and ring-down agree with these values; energy ledger closed (README §1)
At neutrino size	to last as long as the SN1987A neutrinos (proper time 5.3×10⁴ s, Q = 8×10¹⁸) the loop needs an effective vacuum permittivity of 3.4×10¹³ at a = 1.5×10⁻¹² m (Part 2)
Not in the circuit	mass and the weak interaction, by which every neutrino has been detected

The bench ring (a = 10 mm, εr = 100) as the loop kronspice solves, with the values from run_output.txt; f0 and Q recomputed from them.

Every number below was computed in kronspice with an analytic twin beside it and a closed energy ledger.

The idea

The fdamador neutrino is the toroidal photon electron (Williamson and van der Mark, 1997) with its field inverted: the electric field forms the ring and the magnetic field circles round it. It was proposed by Franklin Amador on 4 November 2026 and entered in the kronspice hypothesis register as H28.

In the photon electron and the charged rings before it, a current runs round the ring and B wraps the tube. Here E runs round the ring and B wraps it. The proposal is that this inverted structure is the neutrino, so it is tested against the neutrino's measured properties rather than the electron's.

What Maxwell's equations say before any model

Three results follow from the field equations alone:

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1. **No charge.** A closed loop of E has ∇·E = 0, so by Gauss's law it encloses no charge. It is neutral, as a neutrino is, and so it can never be an electron.
2. **No static form.** E can circulate round the ring only while the magnetic flux through the hole changes (Faraday's law). The ring sustains itself as an oscillation: E round the ring is a ring of displacement current, its B threads the hole, and the changing flux drives E.
3. **An oscillating magnetic moment.** The displacement-current ring is a magnetic dipole whose time average is zero. A time-varying magnetic dipole radiates, so the ring loses energy.

This field pattern already exists in engineering: it is the TE01δ mode of a dielectric ring resonator, used in microwave filters, where the dielectric holds the field in place.

Part 1: the E-ring resonator

A 10 mm ceramic ring with εr = 100 resonates at 2.0606 GHz with Q = 40.5, and kronspice agrees with the formula to four figures by two independent methods.

The ring is a lumped loop of 24 sections: series capacitors for the displacement path along the ring, the ring's self-inductance, a radiation resistance for its magnetic-dipole radiation, and dielectric loss. Radiation alone gives a closed-form Q:

```
Q_{rad} = \frac{6\Lambda}{\pi (ka)^3}, \qquad ka = \frac{\sqrt{2}\,(a/r)}
{\sqrt{\varepsilon_r \Lambda}}, \qquad \Lambda = \ln\frac{8a}{r} - \frac{7}
{4}
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So Q grows as εr^1.5: **the permittivity is the confinement.** Bench ring: a = 10 mm, r = 2.5 mm, tan δ = 1×10⁻⁴.

εr	f_0 (GHz)	ka	Q, formula	Q, kronspice AC
30	3.762	0.788	6.68	6.68
100	2.061	0.432	40.52	40.52
300	1.190	0.249	207	207
1000	0.652	0.137	1,140	1,140

At εr = 100 a ring-down gives Q = 40.53. Its ledger closes: the 3.151×10⁻¹³ J kick equals 3.144×10⁻¹³ J radiated and heated plus 6.7×10⁻¹⁶ J still stored, with a time-stepping error of 2.5×10⁻¹⁸ J. No charge leaves the ring, and the moment averages to −0.007 of its peak. At εr = 30 the ring is no longer small against the wavelength (ka ≈ 0.8), so that row sits at the edge of the lumped model.

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Part 2: the neutrino test bench

To be a neutrino the ring would need an effective vacuum permittivity of 3.4×10^{13}, and nothing known supplies one.

In vacuum ($\varepsilon r = 1$, $a/r = 4$) the ring is electrically large at its own resonance ($ka = 4.3$), so the lumped model fails and gives $Q \approx 0.04$: the field is gone within one period. The field solve of the uninverted photon loop shows the same, with light leaving in under half a period.

The bench asks what holding it would take. SN1987A neutrinos (about 10 MeV) crossed 168,000 light-years. At a declared, illustrative mass of 0.1 eV the Lorentz factor is 10^8, so the proper time is 5.3×10^4 s and the ring must keep its energy for $Q = 8\times10^{18}$.

Property	E-ring model	Measured or standard
Charge	0 ($\nabla\cdot E = 0$)	consistent with 0
Spin ½	crossed inverted ring keeps half-integer modes (Part 3)	½
Static magnetic moment	0 (it oscillates)	$< 2.8\times10^{-11}$ μB (Borexino 2017); standard $\approx 3\times10^{-20}$ μB at 0.1 eV
Lifetime in vacuum	$\approx 3\times10^{-16}$ s	$\geqslant 5.3\times10^4$ s proper time (SN1987A)
Confinement to survive	effective $\varepsilon r = 3.4\times10^{13}$ at $a = 1.5\times10^{-12}$ m	none known
Mass	not given (0.1 eV declared)	< 0.45 eV (KATRIN 2025)
Weak interaction	not in the model	how every neutrino is detected

Part 3: the inverted travelling wave

Crossing the conductors of the inverted ring line still gives half-integer modes, so the spin-½ property of the photon electron survives the inversion.

Swapping E and B in a ring transmission line gives its circuit dual: series capacitors carry E along the path and shunt inductors carry B round it. That is a backward-wave ladder, where the longest wavelength has the highest frequency. Its modes follow $\omega = 1/(2\mathcal{V}(C\cdot Lg)\cdot\sin(\pi k/N))$, with k whole on the plain ring and half-integer on the crossed one. Built in kronspice with 100 sections, C = 1 nF and Lg = 1 μH:

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Mode	Plain ring (MHz)	Wavelengths	Crossed ring (MHz)	Wavelengths
1	80.11	1	160.21	½
2	40.08	2	53.42	1½
3	26.74	3	32.07	2½
4	20.08	4	22.93	3½

The AC peaks match the ladder formula to within 5×10^{-5}, and both AC ledgers balance.

Part 4: an inflow that balances the outflow

The zero-point field balances the ring's radiation exactly, so the ring never decays to zero, but what it holds is half a quantum of random noise.

By the fluctuation–dissipation theorem (Callen and Welton, 1951), the resistance that radiates the ring's energy is also a noise source: the field it radiates into drives it back. At the neutrino ring's 0.1 eV, thermal noise is negligible even in the 2.7 K cosmic background (occupation e^{-430}), leaving the zero-point part. Inflow and outflow are both booked on the magnetic-field channel, so they share one ledger column. Results from a scaled loop at the bench ring's radiation Q of 40.7, over 4 realizations:

Quantity	kronspice	Expected
Stored energy	$0.4941 \pm 0.0067\ \hbar\omega_0$	$0.4916\ \hbar\omega_0$ ($\frac{1}{2}\hbar\omega_0$ for the modes present)
Magnetic channel	391.77 J in = 389.36 J out + 2.15 J stored	in = out + stored
Energy fluctuation σ/⟨E⟩	0.97	≈ 1 for a random field
Mean magnetic moment	1.9×10^{-4} of its rms	0

Does the Casimir effect add inflow? No, but the boundaries that cause it change the outflow. The Casimir force comes from walls changing which zero-point modes fit between them, a fixed energy shift rather than a continuous supply. The same walls change the ring's radiation rate by a Purcell factor F, and the noise changes with it, so the held energy stays put:

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Purcell factor F	Q	Stored, kronspice ($\hbar\omega_0$)	Expected ($\hbar\omega_0$)	Lifetime
4 (enhanced)	10.2	0.4794 ± 0.0023	0.4817	×¼
1 (free space)	40.7	0.4941 ± 0.0067	0.4916	×1
0.2 (suppressed)	203	0.4843 ± 0.0318	0.4942	×5

A coherent ring lasting SN1987A's proper time would need $F \approx 5\times10^{-21}$. The classic measurement of inhibited emission found about 20× (Hulet, Hilfer and Kleppner, 1985), and a free neutrino has no walls.

What the expected value caught. The first version read 8 % high at low Q. The noise started at a non-zero voltage, so the start-up operating point charged the series capacitors through 1 GΩ bleeders, an offset that relaxes over 4×10^{9} s and biased the stored energy in proportion to R_rad. Ramping the noise on from zero removed it, and both points then agreed within 0.5σ. The ledger itself stayed closed, because the offset was booked as energy stored at the start; the expected value computed beside the result is what exposed it.

Because the ramp was added after the mismatch was seen, it was tested for whether it steers the result: on identical noise, ramps of 2Q, 5Q and 8Q agree to four digits, and an abrupt start reproduces the bias exactly. The full account, with the equations, the record of events and the check, is in Reference: the 8 % zero-point bias.

Part 5: implosion and explosion

The ring has both halves, but the implosion must be supplied from outside: only the exact time reverse of its ring-down is taken in whole (0.99966), and then it explodes as the ordinary ring-down. This tests H30 (a particle as a cycle of contraction and expansion) on the E-ring. In standard physics the inward half of a radiating resonator is its ring-down run backwards, an incoming wave growing as e^{+αt} at f_0; every joule of it is declared and booked.

Circuit (kronspice). The Part 1 bench ring (2.0606 GHz, Q 40.52), with its radiation resistance R_rad treated as a port to the outside. Every drive carries the same incident energy, 2.298×10^{-10} J, over 8 τ_E ($\tau_E = Q/\omega_0 = 3.129$ ns). "Absorbed" is the share not sent back while the drive lasts.

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Incoming wave	Absorbed	Peak stored / incident	Explosion / incident
Time-reversed ring-down (coherent perfect absorption rate)	0.999656	0.99536	0.99123
Same shape, twice the rate	0.889452	0.88742	0.88373
Same shape, half the rate	0.901874	0.89483	0.89112
Constant burst at f_0, 2 τ_E	0.799747	0.79681	0.79350
Constant burst at f_0, 8 τ_E	0.486970	0.47658	0.47459
The ring-down itself, not reversed	0.028366	0.53939	0.02031

- The time-reversed drive's whole reflection, 3.4×10^{-4}, is its switch-on (it starts at e^{-8} of its final power). This is coherent perfect absorption, the time-reversed laser (Chong, Ge, Cao and Stone, *PRL* 105, 053901, 2010).
- Ledgers close to 6.7×10^{-6}. CPA run: incident 2.29791×10^{-10} J = reflected 7.914×10^{-14} J + explosion 2.27775×10^{-10} J + heat 1.857×10^{-12} J + stored 7.77×10^{-14} J.
- **Pulsing ring.** With one implosion every 40 periods, the steady cycle takes in 2.175×10^{-10} J and gives back 2.158×10^{-10} J. The difference, 1.718×10^{-12} J (0.79 %), is the dielectric heat (1.716×10^{-12} J). A drive in antiphase settles to the same cycle by cycle 4.

Field (em/fdtd.go). The same torus (εr = 100, lossless) in the field solver, inside a closed Huygens surface at ±2a. The surface feeds in the time reverse of the ring's own outgoing field, growing at the ring-down rate, then switches off at an E maximum.

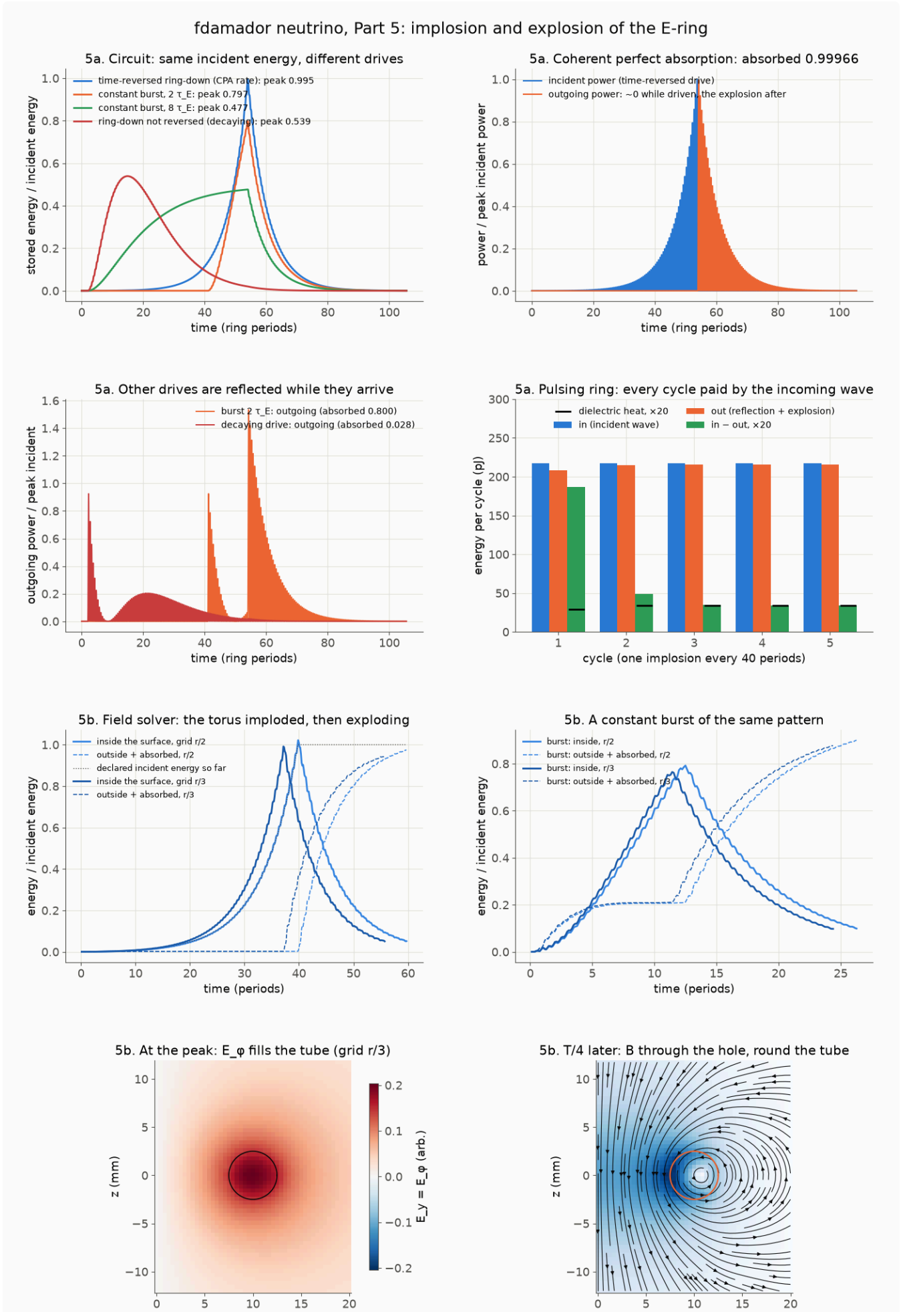
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Quantity	Grid r/2	Grid r/3	Circuit
f_0	2.101 GHz	2.0571 GHz	2.0606 GHz
Q	41.69	38.9	Q_rad 40.68
Held at the peak / delivered	0.9976	0.9974	0.99536 of the incident
Constant burst, peak held	0.7923	0.7649	0.79681
E round the ring: share of the electric energy at the peak	0.9966	0.9971	1
Correlation with the free ring-down's E	0.9991	0.9996	—
Field ledger residual / incident	3.3×10^{-13}	4.8×10^{-13}	5.1×10^{-6}

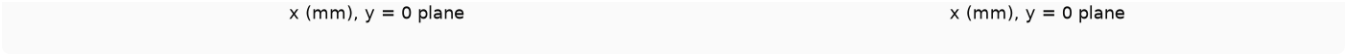
The lumped f_0 and Q lie between the two grids (f_0 moves 2 % and Q 7 % between them; a finer r/4 grid was not run). At the peak the rebuilt field is the TE01δ pattern: E round the ring, and a quarter period later B through the hole, closing round the tube. After the peak 0.947 of the delivered energy explodes outward within 3 τ_E.

Standard prediction, borne out by both: the implosion is the explosion reversed, absorbed without reflection only when it is the exact time reverse, and every cycle is paid for by the incoming wave. Nothing in the ring supplies that wave; for a neutrino it would have to arrive from outside, coherent, every period.

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Part 5 in kronspice and the field solver: the same incident energy under different drives; coherent perfect absorption (incident in, nothing out until the explosion); the pulsing ring's books per cycle; the torus imploded and exploding on grids r/2 and r/3; at the peak E fills the tube round the ring, and a quarter period later B threads the hole.

Part 6: flavours as wave types (H39)

The three wave types are stable states of the ring and, with a defect added, they convert into one another in the same form as neutrino oscillation; tuned, the ring reproduces the measured mixing angles, but that is a fit, and it has nothing of what defines a flavour. The hypothesis, entered as H39 on 2026-11-07: the three neutrino flavours are three kinds of wave on the ring, constructive, destructive and standing.

Where the types live. The uniform E-ring (TE01δ) is a single state with no travelling or standing form. The types live on the ring's harmonics that vary round it, built as the inverted ring lines of Part 3 (24 sections). Each harmonic is a pair of waves running opposite ways, A (clockwise) and B (anticlockwise). One fixed definition, taken at a reference point c:

Type	Definition	What it means
C, constructive	travelling, B = 0	every point reaches the full amplitude
S, standing	|A| = |B|, antinode at c	fixed nodes and antinodes
D, destructive	|A| = |B|, node at c	the two waves cancel at c

No state cancels everywhere: the energy is $N(|A|^2 + |B|^2)$ exactly, so only zero does. And $C = (S + iD)/\sqrt{2}$: **the three types span two dimensions, not three.**

Test 1, persistence. Each type, imposed on a perfect ring and followed for 100 periods in kronspice, keeps its own share to 1.0000000000 on both the plain and the crossed ring. C turns once a period (0.99909 turns), S and D keep their nodes, and the energy falls only by the shunt loss (ledgers about 10^{-4}). None decays or converts.

Test 2, conversion. A capacitor defect δ on one section splits the pair, and a wave started clockwise turns anticlockwise and back: $P = \sin^2 2\theta \cdot \sin^2(\pi \Delta f \cdot t)$, the form of two-flavour oscillation.

Defect δ	Δf, exact	Δf, kronspice	$\sin^2 2\theta$
0.05	37,428 Hz	37,327 Hz	1.0000

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Defect δ	Δf, exact	Δf, kronspice	$\sin^2 2\theta$
0.10	71,454 Hz	71,266 Hz	1.0000
0.20	131,000 Hz	130,660 Hz	1.0001
0.40	224,530 Hz	223,990 Hz	1.0001

Started as a standing wave s sections from the defect, the angle is $\theta = \kappa \cdot s$: kronspice gives $\sin^2 2\theta = 0.2507$, 0.7512 and 1.0000 against 0.25, 0.75 and 1.

Three flavours need a third state. The ring's band-edge π-mode lies 0.86 % below the pair, and two defects mix the three. Tuned for the closest match to the measured mixing (NuFIT 6.0, normal ordering with Super-Kamiokande atmospheric data):

Quantity	Ring, tuned	Measured
Defects	δ = 0.2064 on section 2, 0.1628 on section 23	—
θ_{12}	33.66°	33.68° ± 0.72
θ_{23}	42.28°	43.3° ± 0.9
θ_{13}	8.56°	8.56° ± 0.11
Splitting ratio	33.56	$\Delta m^2_{31}/\Delta m^2_{21} = 33.55$
CP phase	0 or 180° only	212° (+26/−41)

That is four numbers fitted with four knobs, not a prediction. Up to 8.7 % of each mode lies outside the three ring states; kronspice follows the ring's full 24-mode expansion to 1.1×10^{-4} but departs from the 3×3 formula by up to 0.277.

Test 3, implosion and explosion of the coherent unit. A port on the ring at the reference point, with the same external Q as the E-ring (40.5):

Type	Free explosion: energy left	State left	Rebuilt by the reversed explosion: fidelity
S (antinode at the port)	0.0000	—	1.00000 (one port)
D (node at the port)	0.9957	D	cannot be built through this port
C (travelling)	0.4979	D	1.00000 (two ports in quadrature); 0.50000 (one port)

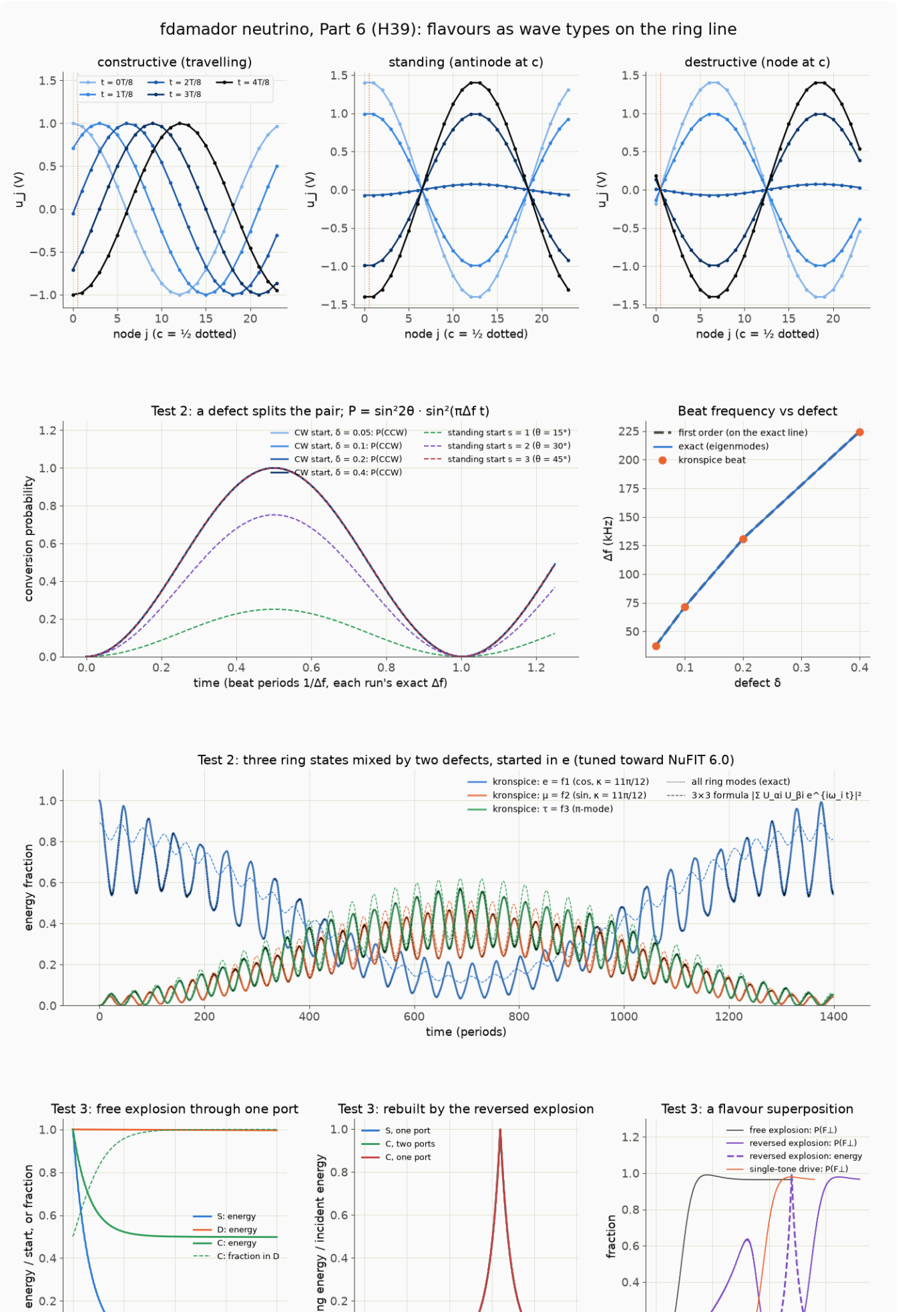
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Type	Free explosion: energy left	State left	Rebuilt by the reversed explosion: fidelity
Flavour mixture, θ = 22.5°	0.0221	the beat continues	0.98440 (a single tone: 0.96903)

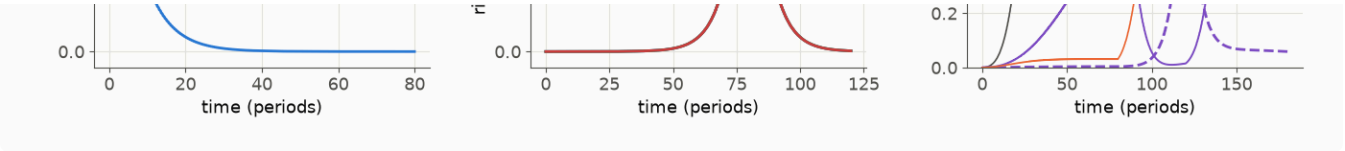
- The explosion changes the type: a standing wave with its antinode at the port leaves completely, one with its node there cannot leave, and a travelling wave loses half and is left standing.
- The implosion rebuilds any type it can reach, but only when the incoming wave is the exact reverse; a travelling wave needs two ports, and a flavour mixture needs both mode frequencies in the right phases.
- A flavour mixture keeps beating through the explosion, and the beat resumes after the rebuild.

Standard prediction: every combination of a degenerate pair is stationary, and the oscillation formula is the generic mathematics of coupled modes. What does not match: a neutrino's flavour is set by the weak interaction (the charged lepton it makes or is made with), its splittings are mass differences entering as $\Delta m^2 L/4E$, matter changes them (MSW), and a CP phase may be present. The ring has no weak production or detection, no mass, no matter effect, and a reciprocal lossless ring cannot carry a CP phase. H39 is recorded as *bounded*.

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Part 6 in kronspice: the three wave types on the ring; the beat a defect makes, against δ; three ring states mixed by two defects tuned toward NuFIT 6.0 (kronspice against the full mode expansion and the 3×3 formula); each type's free explosion, its rebuild by the reversed explosion, and a flavour mixture through the cycle.

Part 7: a spinning ring

After Part 6 the ring still lacks a CP phase (its mixing matrix is real, so δ_CP is 0 or 180°), the neutrino's left-handedness, spin ½, mass and confinement. Part 7 asks whether spinning the torus, about its axis or around the E ring, supplies any of them. "Spinning" has two readings, tested separately: a material ring that turns (7a, in kronspice) and a field that circulates (7b, in the field solver).

7a, a ring that turns. A turning ring delays the co-rotating wave (the Sagnac effect), splitting each resonance by $\Delta\omega/\omega = 2a\Omega \cdot v_g/c^2$. In the ring's own frame this is a Coriolis term, which does no work; in kronspice it is a gyrator across every section of Part 6's ring line.

Check	Result
Splitting, kronspice against exact	agrees to 3.8×10^{-6} (plain and crossed ring)
Work done by the rotation on the waves	6.4×10^{-19} of the stored energy: none at constant Ω
Defect plus rotation, mixing $\sin^2 2\theta$	0.5030 against 0.5000 predicted; any angle is now possible (Part 6: only 45°)
CP phase, δ_CP free within the measured band	188°, J = −0.0050, measured angles kept
CP phase pinned at the measured 212°	211°, but θ_{13} 8.21° (−3.2σ) and splitting ratio −6.5 %
T violation, max |P(e→μ) − P(μ→e)|	0.173 (exact modes 0.173); 0.013 without rotation, the readout's own

So the rotating ring does violate T, and so CP. The catch is speed. On the 10 mm bench ring, the 188° point needs a rim speed of 2.1×10^4 m/s and the 211° point 8.3×10^4 m/s, even with the most favourable wave speed ($v_g = c$). The strongest carbon-fibre hoop bursts at 1,881

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m/s, so that is 11 to 44 times too fast; the εr = 100 ring needs ten times more again, and the circuit line's own slow wave would need more than the speed of light. At the burst speed δ_CP moves only 0.8° from 180°.

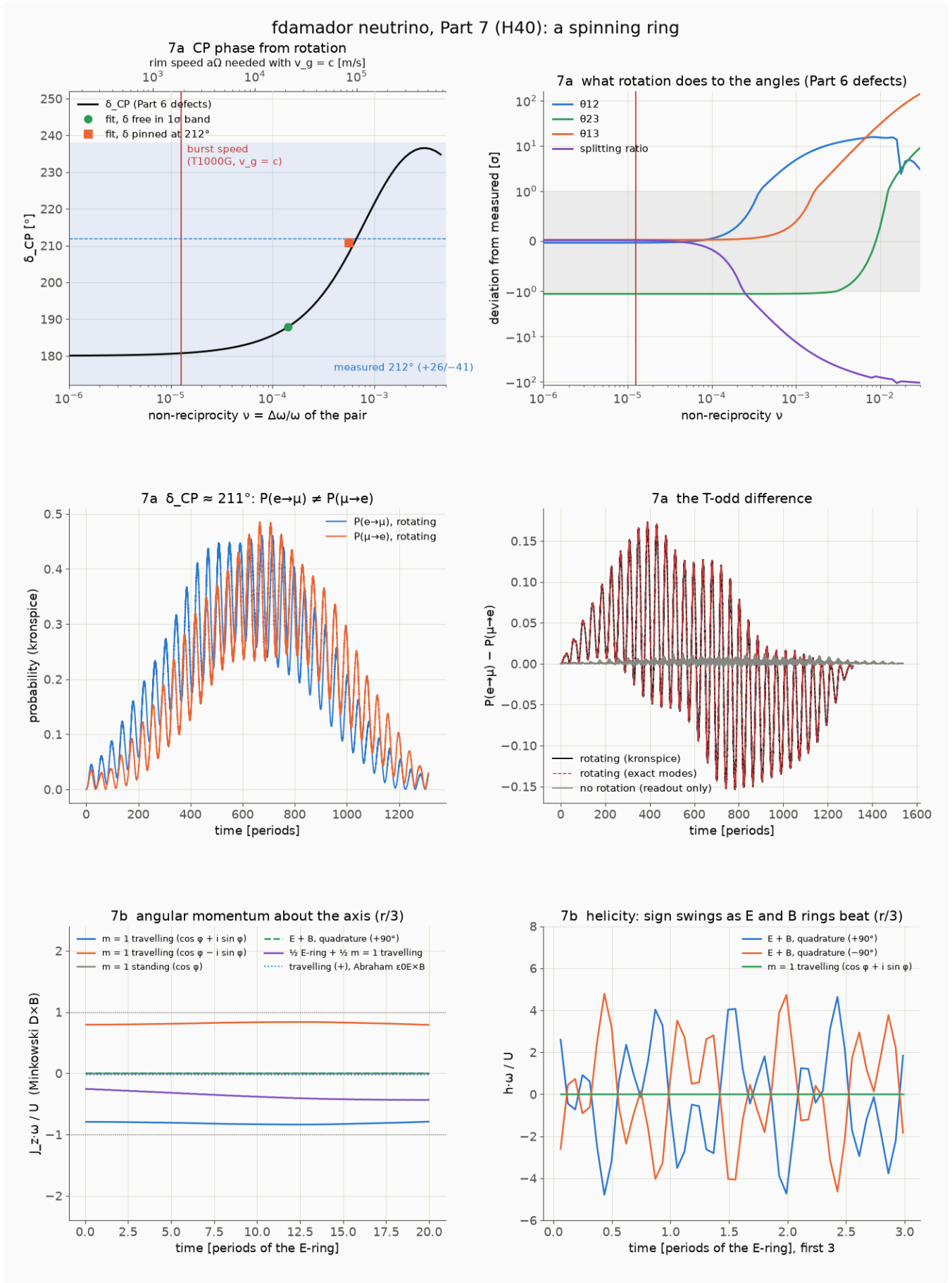
7b, a field that spins (field solver, grids r/2 and r/3; field ledgers closed to 10^{-12}).

- **E-ring plus B-ring (spin about the axis):** energy circulates round the axis, but the total angular momentum about it is exactly zero, because the field looks the same at every angle. Its handedness (∫E·B) starts with the sign of the relative phase, then swings through both signs as the two modes beat: it is not conserved in the dielectric.
- **A turning m = 1 pattern:** it carries angular momentum of a chosen sign and holds it while it radiates. On the finer grid, where the mode is 95.6 % m = 1, it carries 0.79 of one unit per quantum (Minkowski form): it tends to a whole unit, not ½. Spin ½ can only come from a mixture of modes.
- **Decay:** the turning and standing patterns ring down with the same Q, so spinning gives no confinement.
- **Handedness:** Maxwell's equations favour neither sign; nothing selects the left-handed one neutrinos show.

7c, mass. Spinning at the burst speed adds $½v^2 = 1.77\times10^6$ J/kg, a mass change of 2×10^{-11}, paid by whatever turns the ring.

Standard prediction: rotation non-reciprocity is Sagnac physics, as in ring-laser gyroscopes; CP violation is a phase of the weak-interaction (PMNS) matrix; neutrino left-handedness comes from the V−A weak interaction; field helicity and angular momentum are ordinary electromagnetism. H40 is recorded as *bounded*.

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Part 7: δ_CP against the non-reciprocity and the rim speed it needs, with the burst speed marked; what rotation does to the mixing angles; P(e→μ) and P(μ→e) at δ_CP ≈ 211° and their difference (kronspice against the exact modes, and without rotation); angular momentum per quantum of each field pattern; the E + B handedness swinging as the modes beat. The helicity panel uses h·ω/U, which inside εr = 100 can reach 5.05.

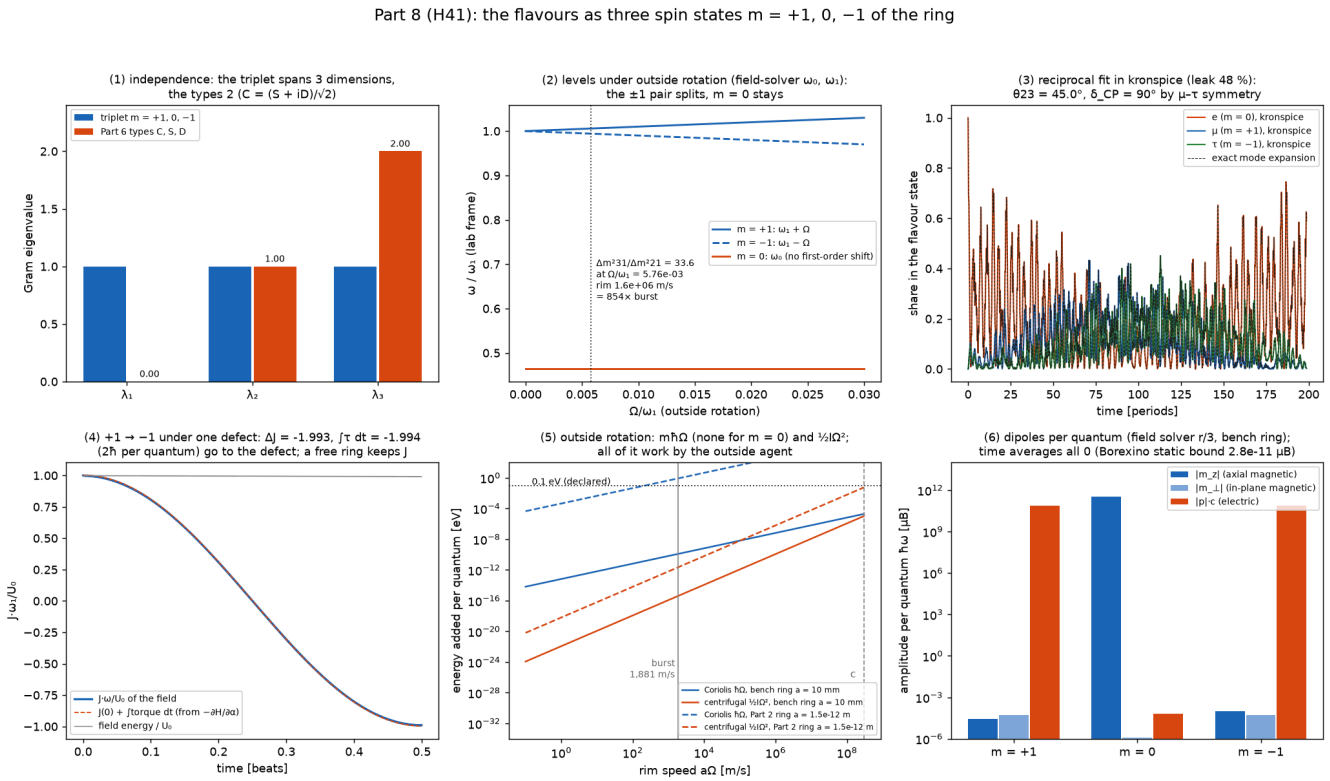
Part 8: flavours as three spin states

The user's proposal: at detection a neutrino is either turning one way with full magnetic field, not turning, or turning the other way. On the ring these are the azimuthal states m = +1, 0 and −1, labelled by their angular momentum J_z, which persists, rather than by the instantaneous B, which passes through zero twice a period in every mode.

Test	Result
8a, independence	Gram eigenvalues 1, 1, 1: three genuine states (Part 6's types gave 0, 1, 2, only two). Field solver: m = 0 at 2.057 GHz, m = ±1 at 4.435 GHz
8b, mixing	Series defects never mix m = 0; a shunt deformation does. With e = m 0 the reciprocal ring forces θ_{23} = 45° (1.9σ) and δ_CP = 90° or 270°; the other angles fit exactly with 4 knobs for 3 targets, putting 48 % of each state outside the triplet. Rotation needs 414× the burst speed
8c, angular momentum	+1 → −1 gives 2ħ per quantum to the defect (field ΔJ −1.993 against the torque on the defect −1.994); with no defect J is held to 3.5×10^{-11}. A free ring cannot switch
8d, magnetic moment	No state has a time-averaged moment (Borexino met trivially). The turning states hold half their energy as magnetic at every instant but carry no magnetic dipole; what turns is an electric dipole. m = 0 carries the axial moment, through zero twice a period
8e, outside rotation	Coriolis adds mħΩ, nothing for m = 0. 0.1 eV needs Ω = 1.52×10^{14} rad/s: 5,100c at the bench rim; 222 m/s on the SN1987A-sized ring, but there Ω equals the ring's own ω and no such material ring exists. Centrifugal energy stays below ½ħω; the measured mass ratio needs 854× the burst speed

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Standard prediction: all three flavours are left-handed, so spin direction separates neutrino from antineutrino, not flavours; flavour is the charged lepton at detection; oscillation keeps the helicity. H41 is recorded as *bounded*.



Part 8: independence of the triplet against Part 6′s types; levels under outside rotation; the reciprocal flavour fit in kronspice; the angular momentum handed to the defect; the energy outside rotation adds against the declared 0.1 eV; the dipoles of each state. Structure figures of each state: viz/structure_m+1.png, structure_m0.png, structure_m-1.png.

Part 9: two neutrinos

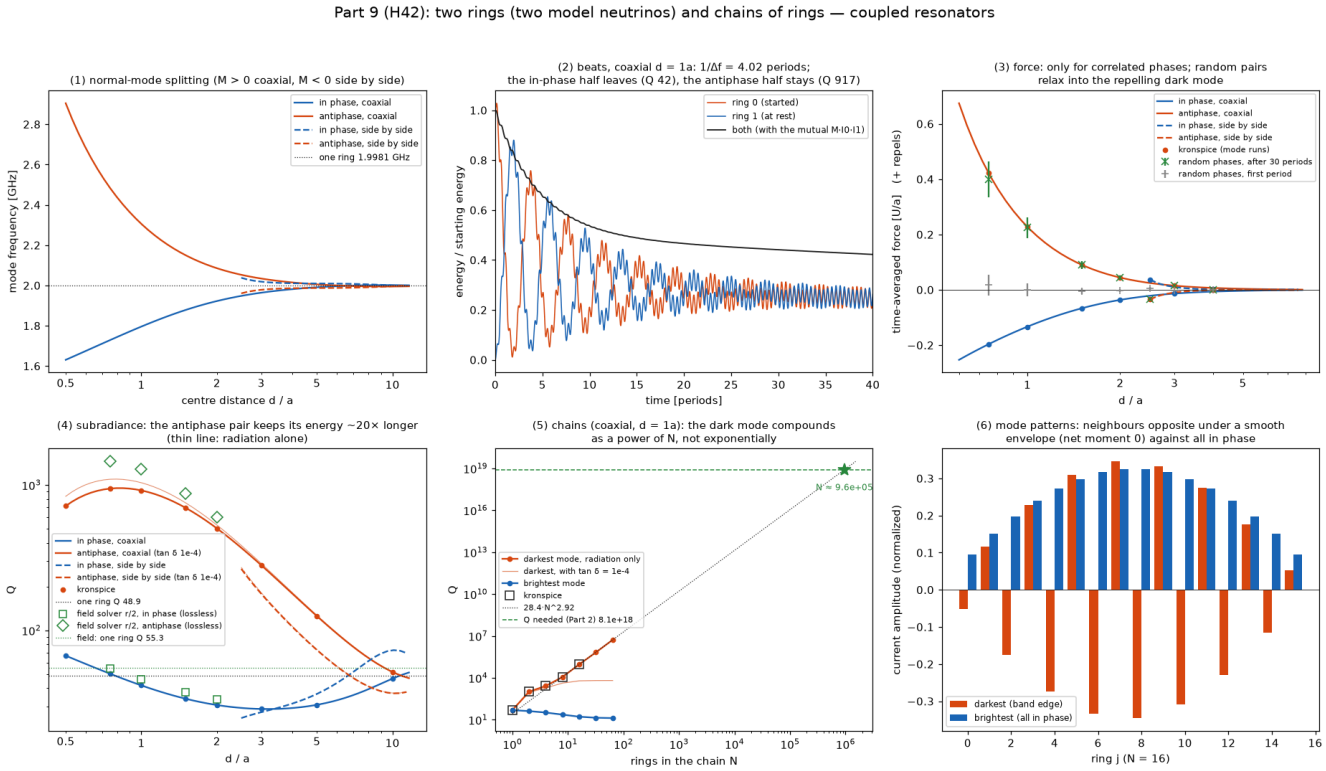
Two identical rings a distance d apart couple through the vector potential: a mutual inductance and a mutual radiation resistance from the same retarded kernel, solved in kronspice and, for the radiation, in the field solver. With the self terms from the same kernel, one ring here is 1.998 GHz with Q 48.9 (Part 1′s small-loop formula gives 2.06 GHz, Q 40.5).

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Test	Result
9a, coupled modes	Coaxial at d = a: in phase 1.796 GHz (Q 41.8), antiphase 2.306 GHz (Q 917); energy beats between the rings at the predicted 535 MHz ($+4\times10^{-4}$) and leaves the energy in the antiphase pattern
9b, force	Coaxial in phase attracts, antiphase repels (side by side the signs reverse); random phases give no force at first, but the pairs relax into the repelling antiphase mode
9c, subradiance	The antiphase pair lives about 22× longer than one ring (circuit ×21.1, field solver ×22.1–23.1 at d = a; best ×26 at 0.75a)
9d, chains	The darkest (band-edge) mode's Q grows as N^2.92 (5.3×10^{6} at N = 64); Q = 8×10^{18} would need about 9.6×10^{5} phase-locked lossless rings; with dielectric loss the chain stops near Q 6,500, and rings in vacuum get no subradiance
9e, against measurement	At neutrino size two rings shift each other by 3.3 meV, 10^{5}× a weak contact term; flyby cross-section about 10^{-32} cm² against about 10^{-41} cm² for 10 MeV neutrinos, about 10^{9}× larger

Standard prediction: neutrinos interact with each other only by Z exchange, too weakly to have been measured in a laboratory; their collective flavour conversion in supernovae comes from forward scattering on each other; Pauli exclusion limits how densely they pack. H42 is recorded as *bounded*.

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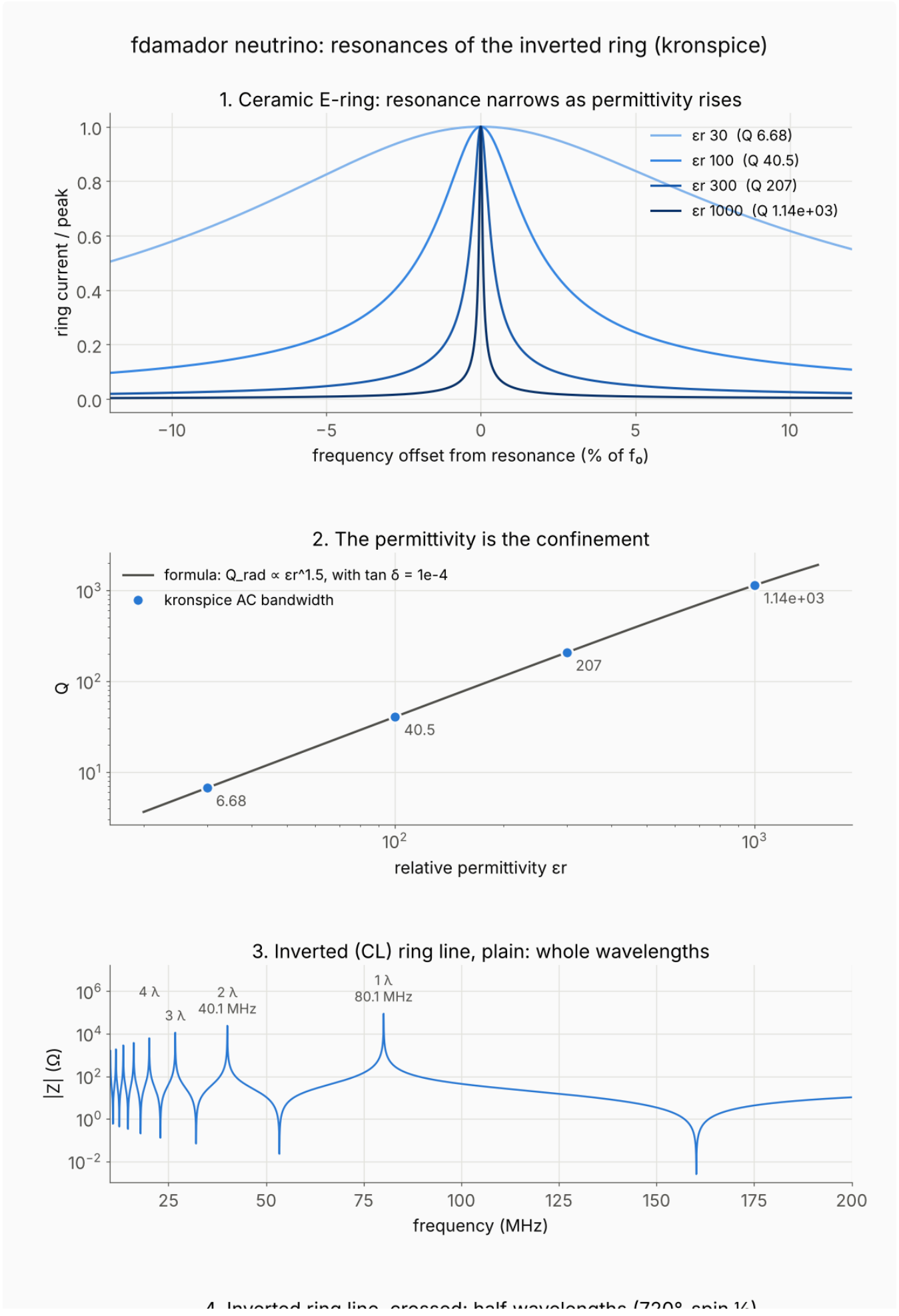


Part 9: mode splitting against distance; energy beating between two rings; the time-averaged force, including random pairs relaxing into the repelling mode; subradiance in kronspice and the field solver; chains, whose dark mode compounds as a power of N; the dark and bright mode patterns. Also viz/structure_pair.png and the animation viz/pair_cycle.gif.

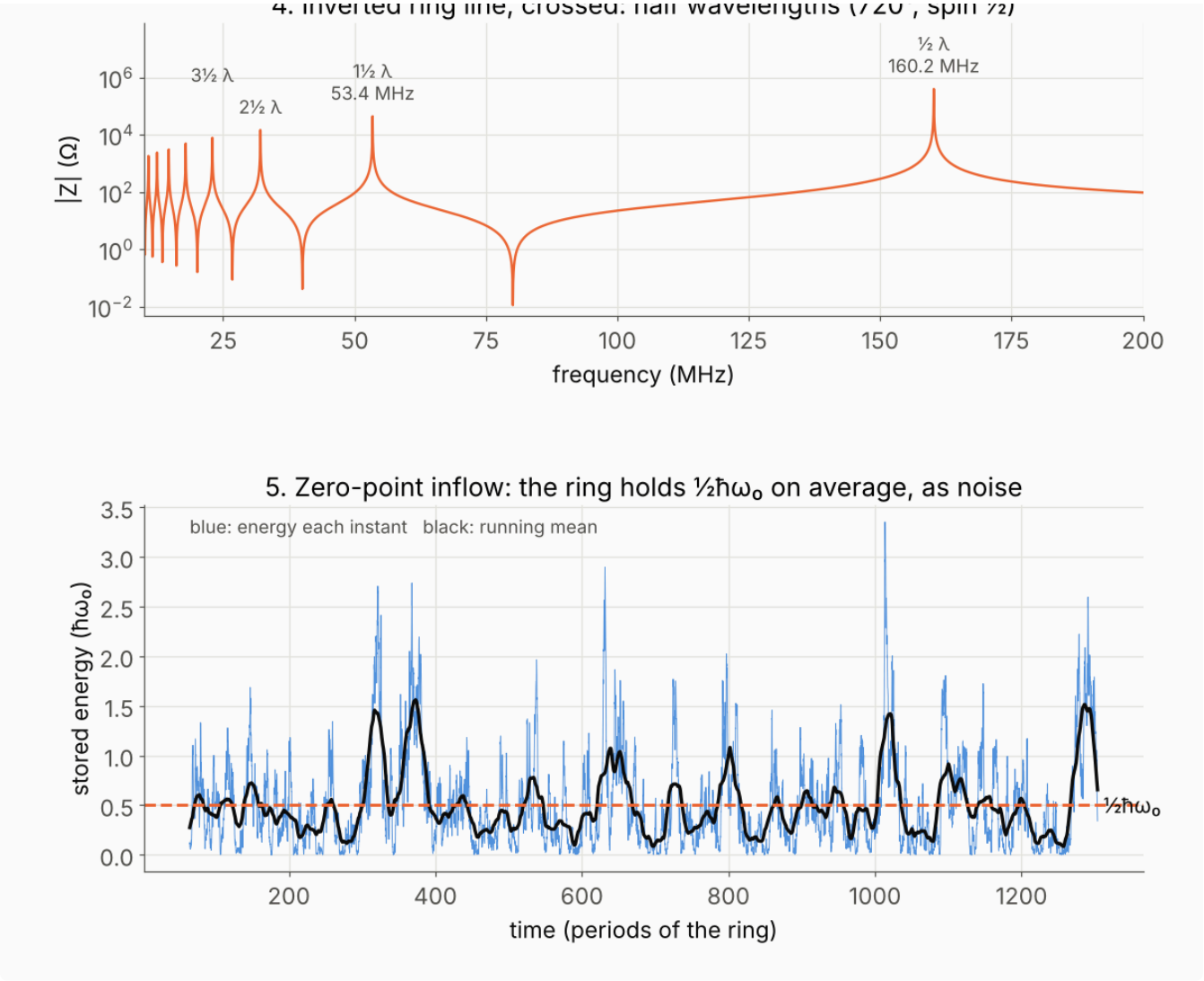
Chart and sound

The resonance chart shows five things: the E-ring's resonance narrowing as permittivity rises, Q on its εr^1.5 line, the plain and crossed ring-line spectra with each mode labelled in wavelengths, and one zero-point run with the stored energy fluctuating about ½ħω₀.

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Every real frequency here is far above hearing (2.06 GHz, 20–160 MHz, 24 THz), so each sound is the kronspice waveform slowed by one factor. Every frequency ratio, decay rate and Q is kept.

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Sound	Source	What you hear
1. Ceramic ring, εr 100 (Q 40.5)	kronspice ring-down	a 0.14 s tick at 440 Hz
2. Ceramic ring, εr 1000 (Q 1,140)	kronspice ring-down	a 3.4 s bell at 440 Hz
3. Ring held by the zero-point field	kronspice noise run	a 2.8 s hiss centred on 440 Hz
4. Plain inverted ring line	four highest modes, synthesized	880, 440, 294, 221 Hz: the undertones 1, ½, ⅓, ¼
5. Crossed inverted ring line	four highest modes, synthesized	880, 293, 176, 126 Hz: the odd undertones 1, ⅓, ⅕, ⅐

The harmonics file adds the ring-line modes one at a time, so each tone can be heard entering. The crossing removes every even member of the series: the spin-½ property heard rather than seen. Sounds 1–3 have no overtones, because each is a single linear resonance.

Conclusions and next steps

The inversion gets the two properties a field picture can get, no charge and no static moment, and keeps spin ½ in its circuit form; it fails on stability and has nothing to say about mass or the weak interaction.

The zero-point inflow closes the energy books but does not rescue the model. It holds every mode of every oscillator at $\frac{1}{2}\hbar\omega_0$, so it singles out no particle, and its fluctuations are as large as its mean.

Parts 5 and 6 add two results. The ring has an implosion as well as an explosion, but only an incoming wave from outside, the exact reverse of its ring-down, supplies it. And the wave types on the ring reproduce the form of flavour oscillation, and when tuned its angles, without the mass, weak interaction or CP phase that make a neutrino's flavour; H39 is recorded as bounded.

Part 7 adds a third. Spinning the ring makes its mixing complex, so it gives a CP phase and T violation in form, but only at rim speeds 11 to 444 times what any real ring survives; a spinning field carries angular momentum of a chosen sign, tending to one whole unit, not ½, and neither form gives left-handedness, mass or confinement. H40 is recorded as bounded.

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Parts 8 and 9 add two more. The three spin states are genuinely independent, but switching between them needs a torque from outside and the measured mixing comes only by fitting; outside rotation cannot supply the mass (H41, bounded). Two rings couple as resonators and an antiphase pair outlives one ring about 22×, but confinement would take about a million phase-locked rings, and the coupling is about 10^9× stronger than the measured neutrino–neutrino interaction (H42, bounded).

H28 stays *bounded* until one of these conditions is met. Each is listed with what has been tried and the result; none is met yet.

- [ ] **Confinement** (radiation suppressed by about 5×10^{-21}): open. Part 4′s zero-point inflow holds only $\frac{1}{2}\hbar\omega_0$ of random noise, not a coherent particle; Part 5′s implosion holds the ring only while an incoming wave from outside feeds it; Part 7′s spinning patterns decay at the same rate as standing ones; Part 9′s subradiance gives about 22× for a pair and N^2.92 for a chain, so about 10^6 phase-locked lossless rings would be needed.
- [ ] **Mass**: open. Part 7c′s rotation adds at most 2×10^{-11} of the ring′s mass at the burst speed of the strongest fibre, paid by whatever turns it. Part 8e′s outside rotation needs $\Omega = 1.5\times10^{14}$ rad/s for 0.1 eV (5,100c at the bench rim), gives nothing to the not-turning state, and its centrifugal energy stays below $\frac{1}{2}\hbar\omega$.
- [ ] **A coupling to the weak interaction**, the only way neutrinos have ever been detected: open. Nothing tried yet; no circuit or field element in the model couples to it.
- [ ] **A prediction that differs from the standard neutrino**, measurable with existing data: open. Parts 6–8 reproduce the form of flavour oscillation and of CP violation only by tuning defects and rotation to the measured values (fits, not predictions). Part 9′s one distinct prediction, rings interacting about 10^9× more strongly than neutrinos, is contradicted by the data.

Repository and sources

The model lives in `examples/fdamador_neutrino` of fdamador/kronspice, on the branch `neutrino-implosion-flavours` (PR #19). All tests pass, including the race detector. Run it with `go run ./examples/fdamador_neutrino` (Parts 1–9; `-only N` for one part, `-field` for the field-solver runs), and draw the charts with `-viz` then the `viz/render*.py` scripts. The electron′s circuit mirror is in `examples/photon_electron`.

Sources:

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- H. B. Callen, T. A. Welton, *Phys. Rev.* 83, 34 (1951) (zero-point noise of a resistance)
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